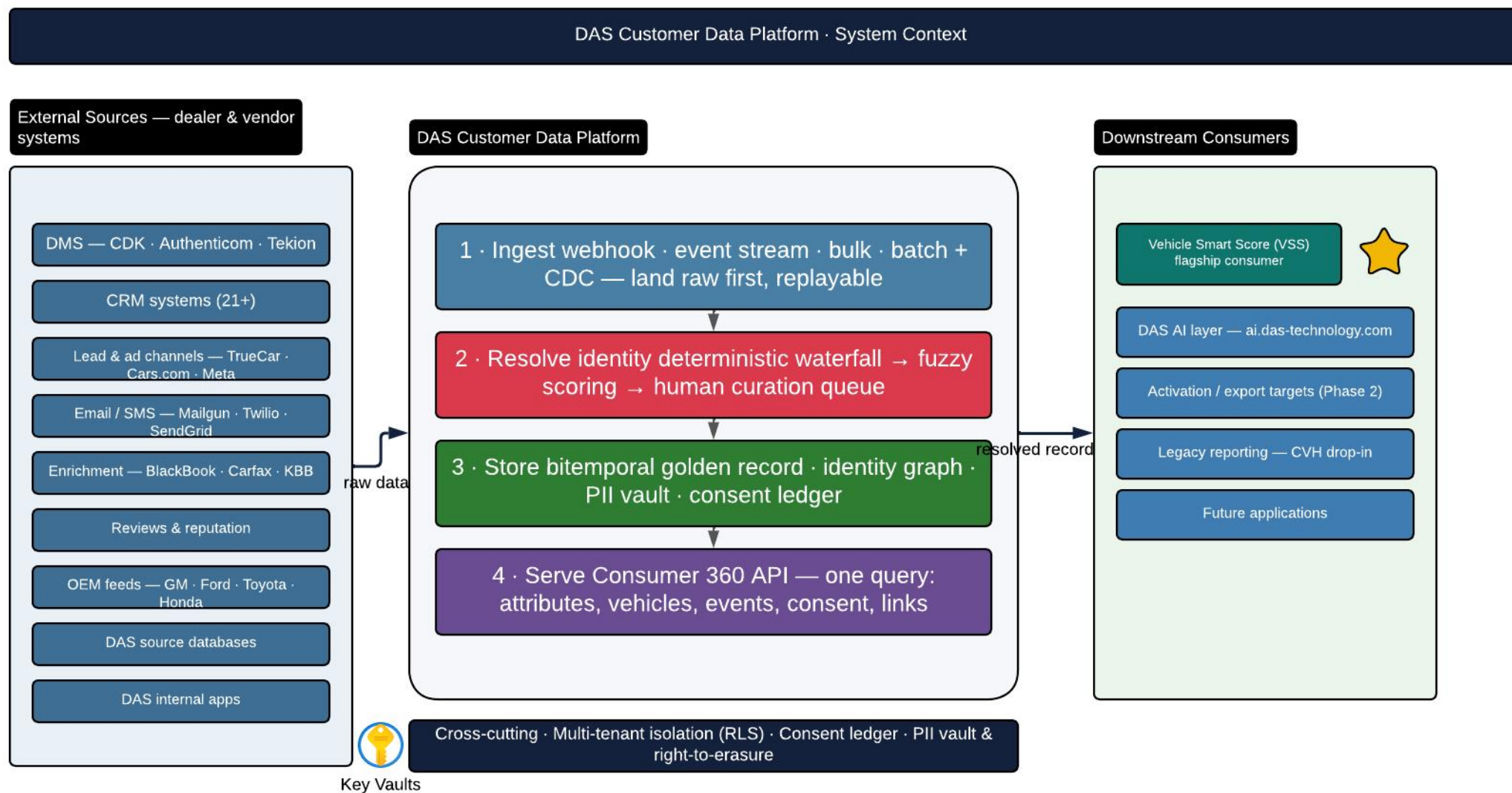
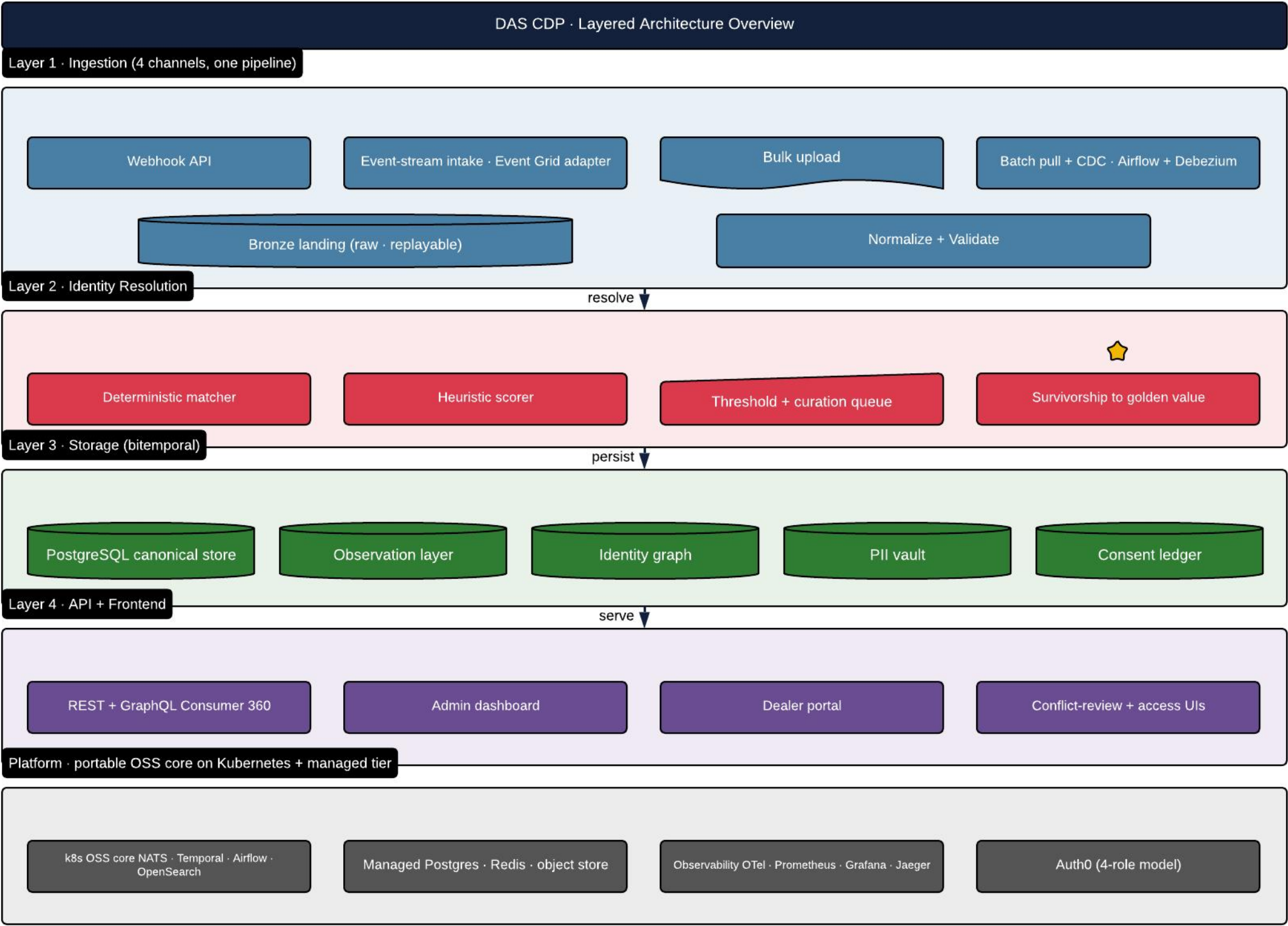


From many dealer & vendor systems to one clean, governed consumer record — resolved once, served to every product.



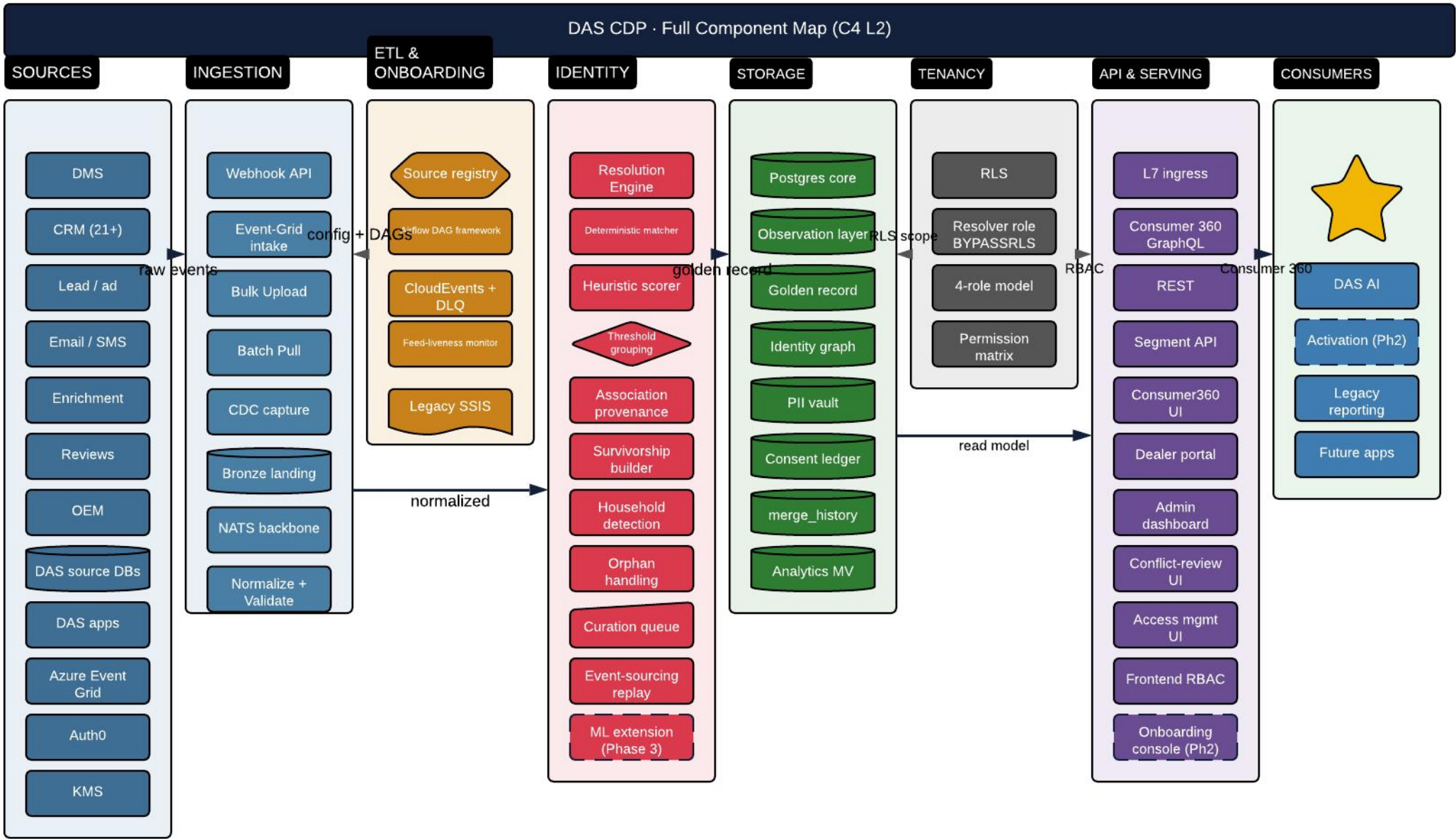
Portable OSS core (PostgreSQL · NATS · Airflow · Temporal · Auth0) — Azure-primary, AWS-portable. Field Catalog v1 = 27 data points, designed to grow. Phase 1 build · Phase 2 activation · Phase 3 AI/ML.

Five layers: four ingestion channels land to a replayable bronze tier · identity resolution emits golden values · a bitemporal Postgres store serves REST/GraphQL and operator UIs - all on a portable OSS core (Kubernetes) plus a managed tier.



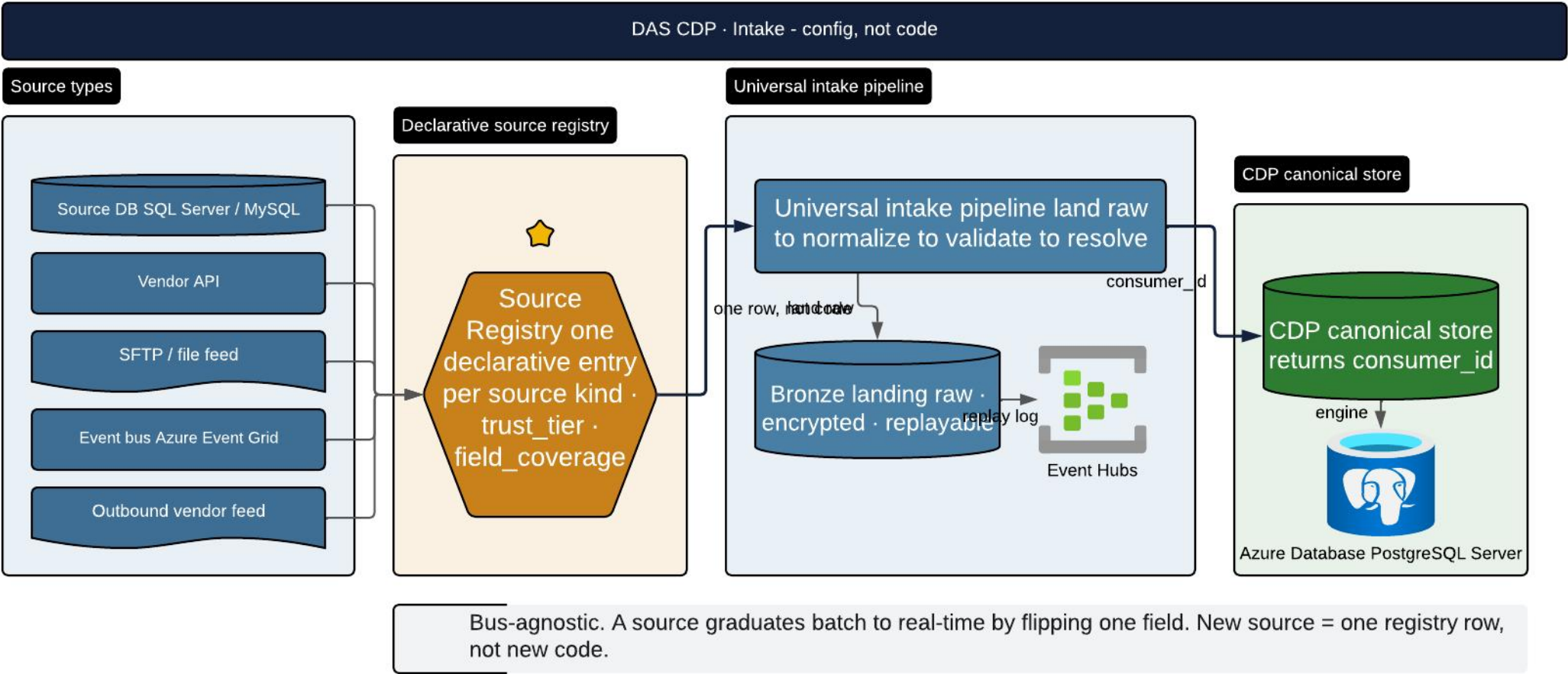
Replaces the prior flat system-design diagram. Portable OSS core on Kubernetes; managed tier maps to Azure today, AWS-equivalent on request. Source: DAS CDP architecture set, 2026-06.

47-component C4 Level-2 map across eight zones; the data spine runs Sources to Ingestion to Identity to Storage to Serving to Consumers, while ETL feeds Ingestion and Tenancy governs Storage and Serving.



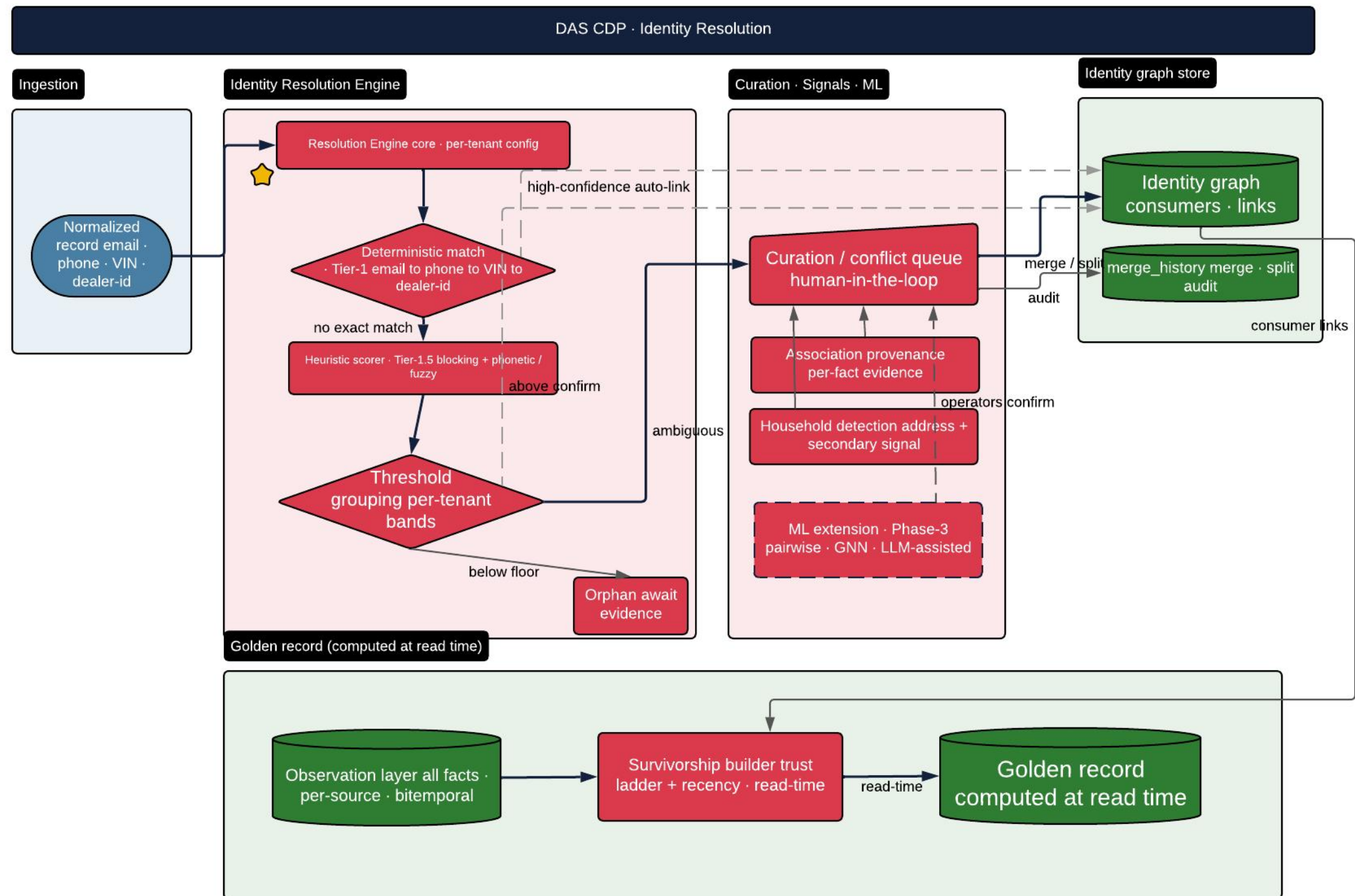
Spine arrows show primary data flow only; ETL connects as feeder, Tenancy as dashed governance. Phase 2/3 items (ML extension, Onboarding console, Activation) shown dashed. Logical view: portable OSS core, no cloud-vendor lock-in (AWS/Azure-neutral).

Every source - DB, API, SFTP, event bus, vendor feed - registers as one declarative entry and flows through a single bus-agnostic pipeline that lands raw, normalizes, validates, and resolves to a canonical consumer_id.



Provenance: docs/diagrams - supersedes the prior intake sketch. Bus-agnostic core (Event Hubs or NATS). Adding a source is a registry row, not a code change.

A tiered engine: deterministic waterfall, then fuzzy scoring with per-tenant thresholds, then a human curation queue - linking every record to exactly one consumer.

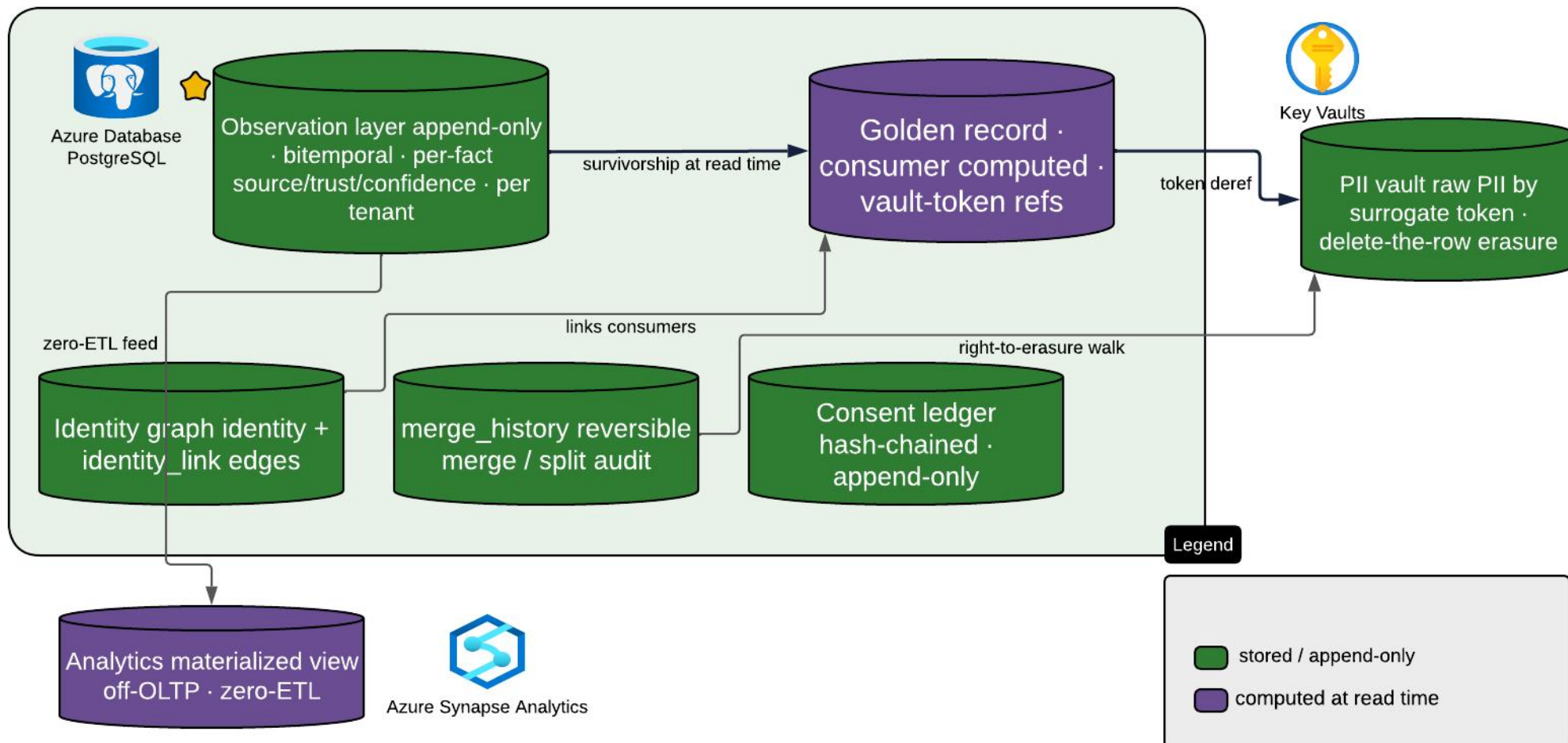


Survivorship is computed at read time (trust ladder + recency), never stored. Phase-3 ML feeds the SAME operator queue - it proposes, operators confirm; it never auto-merges.

A computed golden record: facts land append-only and bitemporal in the Observation layer, survivorship resolves one consumer view at read time, and raw PII lives behind surrogate tokens in a delete-the-row vault.

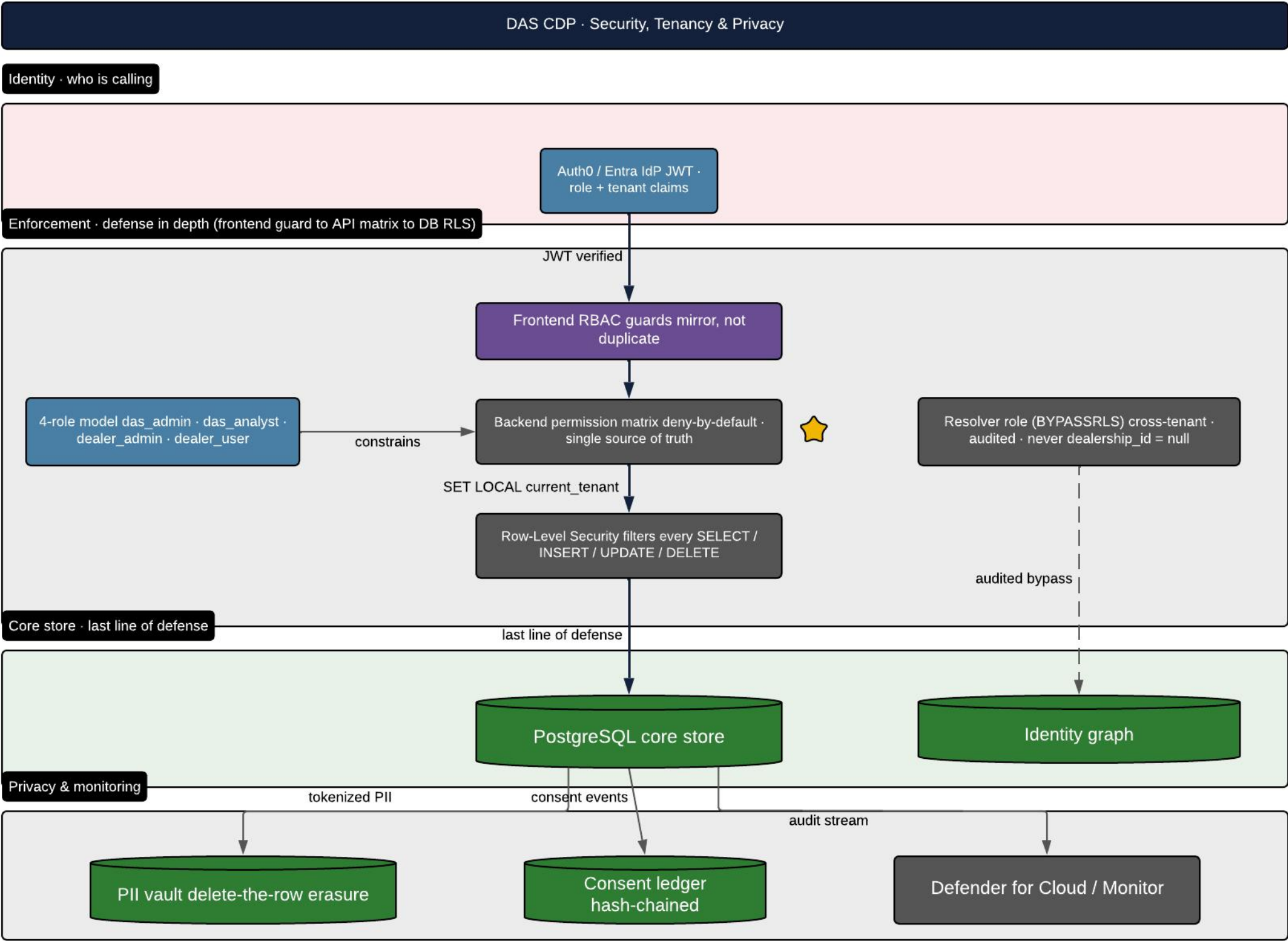
DAS CDP · Data Model & Golden Record

PostgreSQL core store · schema cdp · RLS-enforced



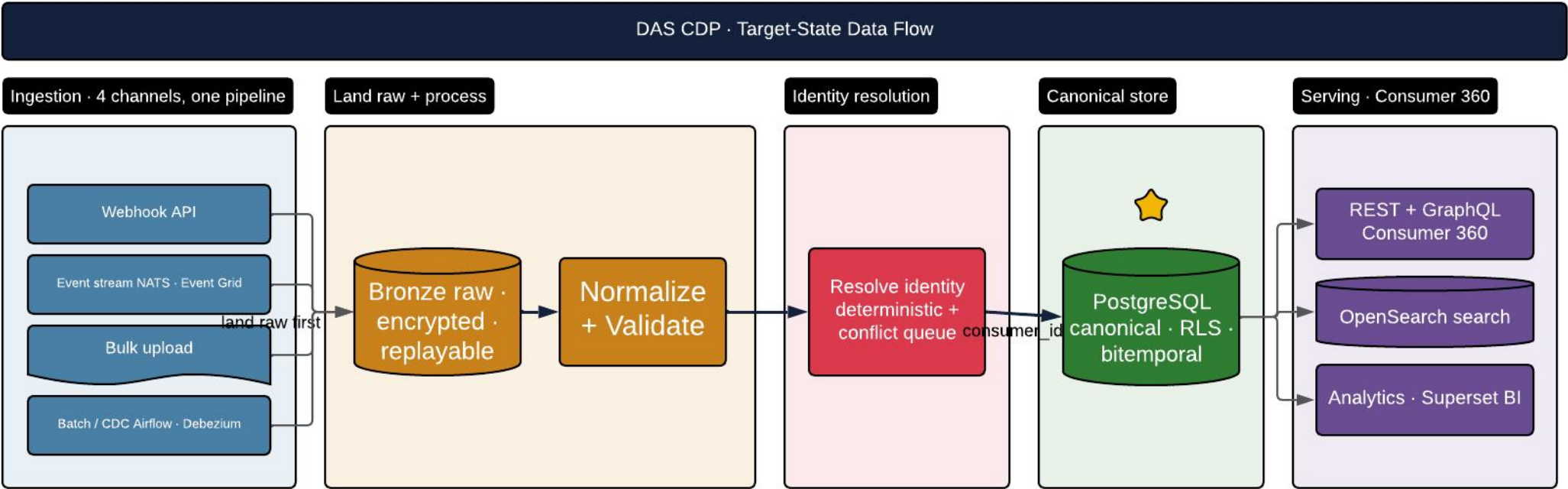
Stored tables are append-only and bitemporal (source · trust · confidence per fact); the golden record and analytics view are computed at read time, never stored stale. Erasure = delete the row in the PII vault; surrogate tokens elsewhere dangle.

Defense in depth, JWT to DB: verified JWT carries role + tenant claims; a deny-by-default matrix is the source of truth; Postgres RLS filters every query as the last line.



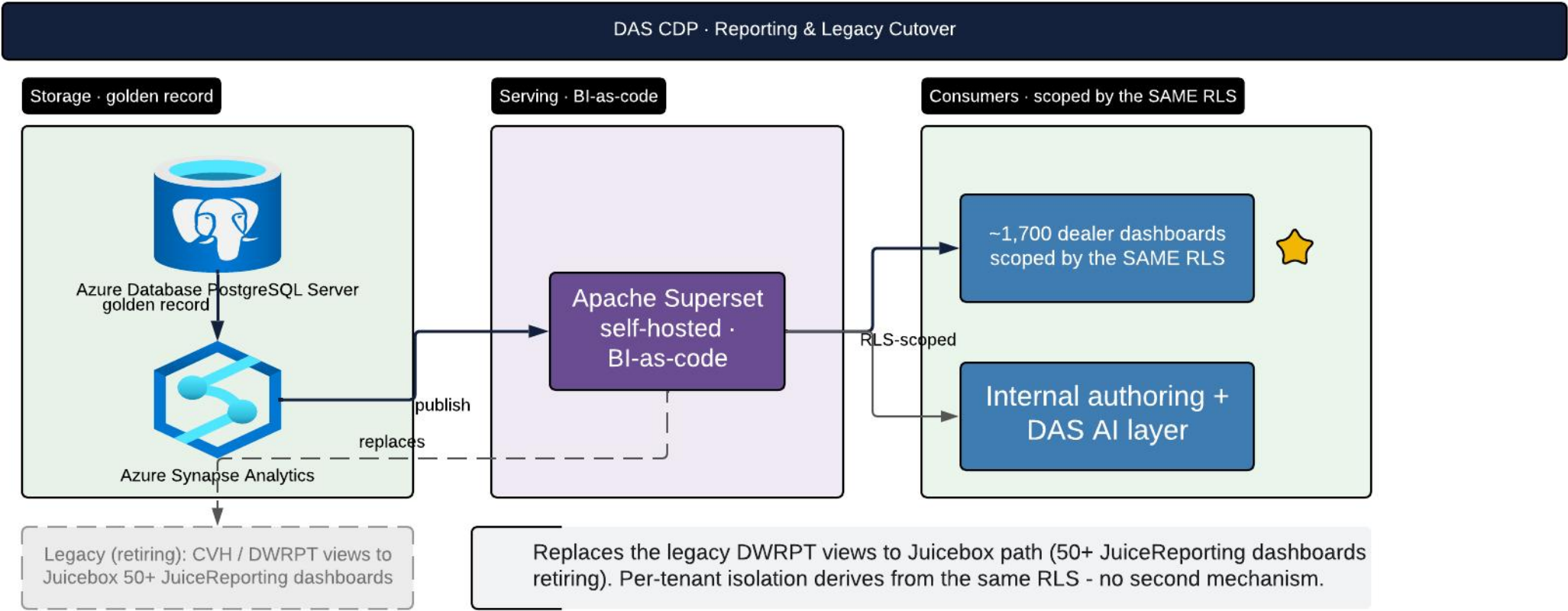
Auth0/Entra IdP to deny-by-default permission matrix to Postgres RLS; resolver BYPASSRLS is the sole audited cross-tenant path. Azure anchor icons illustrative; AWS equivalents apply.

End-to-end runtime path - four ingestion channels land raw in Bronze, then normalize, resolve identity to one consumer_id, and serve a single canonical record.



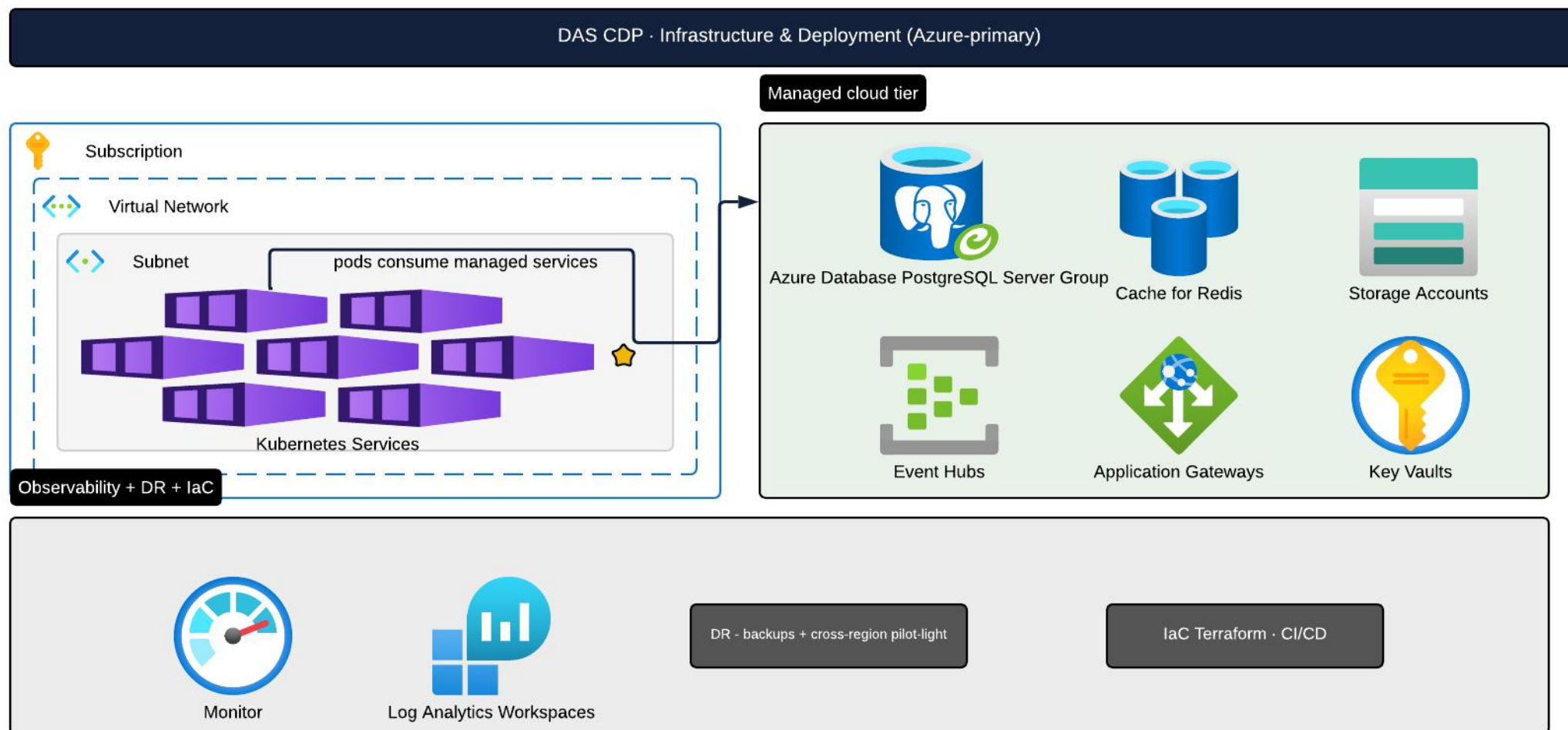
Runtime, left to right: land raw first (replayable), normalize + validate, resolve to one consumer_id, then serve. Portable OSS core; AWS / Azure-equivalent managed tier. Source: docs/system-design.

The CDP becomes the single reporting source of truth: golden record to analytics store to Superset to ~1,700 dealer dashboards, all isolated by the same per-tenant RLS.



Source of truth: PostgreSQL golden record · per-tenant Row-Level Security · Apache Superset (self-hosted, BI-as-code, \$0 license). Supersedes the CVH / DWRPT to Juicebox path.

Azure-primary reference deployment: the portable OSS core runs in AKS inside a VNet, stateful and edge concerns lease managed services, and observability, DR, and IaC cut across - every component maps to an AWS analog in the footer.



Portable OSS core - identical on AWS (EKS · RDS/Aurora Postgres · S3 · MSK · ALB · KMS). Cloud substrate pending AWS-vs-Azure bake-off.